

LAB 3: SOFTWARE ENGINEERING

COMPONENT MODELLING & ARCHITECTURAL PATTERN

SELECTION

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1. Architecture Analysis:

Layered Architecture: The system is divided into layers, where each layer has a specific responsibility. It is simple to understand and organize, but changes in one layer may affect the layers connected to it.

Microservices Architecture: The main functions such as order management, inventory checking, supplier ordering, payment and shipment tracking can be handled as separate services. This makes it easier to develop, update and scale individual services.

Client-Server Architecture: Clients communicate with a central server that handles the main processing. It is simple to manage, but a heavily loaded central server can become a bottleneck.

2. Architecture Selection:

Selected Architecture: Microservices Architecture

Reason 1: The dropshipping system has different responsibilities such as receiving orders, checking supplier stock, creating supplier purchase orders and tracking shipments. These responsibilities can be separated into services, so each part is easier to manage and maintain.

Reason 2: Different parts of the system may have different workloads. For example, order processing may receive many requests while shipment tracking may have a different load. With microservices, a required service can be scaled without scaling the complete system.

3. Security Advantage:

Security checks such as authentication and authorization can be applied at the service level. This helps restrict access to order and supplier information and prevents unauthorized systems from directly using protected services.

4. Performance Benefit:

The Order Manager can receive and validate incoming order webhooks quickly while other tasks are handled by separate services. This reduces unnecessary processing in the initial request and allows individual services to be scaled when the number of orders increases.

5. Main Components:

The component model contains Order Manager, Payment Service, Supplier Inventory, Supplier Order and Shipment Tracking. The components communicate through defined interfaces, while the Storefront Platform and Supplier Partner act as external systems.